Campus Event Financial Planning

Complete worked solution

Calculate fixed venue, equipment, and transport costs plus per-participant catering and certificate costs. Sponsorship is applied first. A registration fee is collected per participant. Compute remaining funding gap after sponsorship and registrations. If the per-person registration contribution exceeds the variable cost per attendee, calculate the minimum whole participant count to cover all costs after sponsorship; otherwise report that registration income cannot break even.

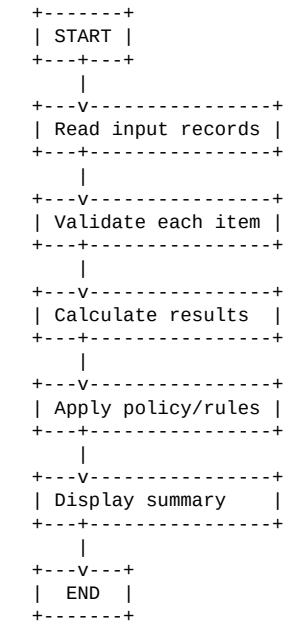
1. Computational Thinking and Algorithm

Inputs: Supplied records and policy values. **Processing:** Validate data, calculate charges or scores, apply ordered rules, and compare outcomes. **Outputs:** Itemized results, classifications, rankings, or recommendations.

Pseudocode

```
START
  Read fixed costs, per-person costs, sponsorship, fee, capacity
  Validate all values
  Calculate fixed cost and variable cost per participant
  For planned attendance, calculate total cost and funding gap
  Calculate contribution per registration = fee - variable cost
  IF contribution is positive
    Find smallest participant count that covers fixed cost after sponsor
    Check against venue capacity
  ELSE report registration alone cannot reach break-even
  Display financial summary
END
```

2. Flowchart



3. Python Program

The program uses basic variables, numeric and string data, tuples, lists, loops, conditionals, arithmetic, and formatted output. Policy values are constants near the top of the program.

```
venue = 85000
equipment = 24000
transport = 18000
```

```
catering_per_person = 650
certificate_per_person = 120
planned_participants = 220
capacity = 300
sponsorship = 70000
registration_fee = 1100

values = [venue, equipment, transport, catering_per_person,
          certificate_per_person, planned_participants, capacity,
          sponsorship, registration_fee]
valid = True
for value in values:
    if value < 0:
        valid = False
if not valid or planned_participants > capacity:
    print("Invalid event inputs or attendance exceeds capacity")
else:
    fixed_cost = venue + equipment + transport
    variable_per_person = catering_per_person + certificate_per_person
    total_cost = fixed_cost + variable_per_person * planned_participants
    registration_income = registration_fee * planned_participants
    total_funds = sponsorship + registration_income
    shortfall = max(0, total_cost - total_funds)
    surplus = max(0, total_funds - total_cost)
    contribution = registration_fee - variable_per_person
    print("Fixed costs:", fixed_cost, "Variable cost/person:", variable_per_person)
    print("Planned total cost:", total_cost,
          "Registration income:", registration_income,
          "Sponsorship:", sponsorship)
    print("Funding shortfall:", shortfall, "Surplus:", surplus)
    if contribution <= 0:
        print("Registration fee does not cover variable cost; break-even is impossible.")
    else:
        remaining_fixed = max(0, fixed_cost - sponsorship)
        # Integer ceiling using arithmetic available at this syllabus level.
        minimum_count = int((remaining_fixed + contribution - 1) // contribution)
        print("Minimum registrations for break-even:", minimum_count)
        if minimum_count > capacity:
            print("Break-even exceeds venue capacity.")
        elif minimum_count > planned_participants:
            print("Increase registrations to at least", minimum_count)
        else:
            print("Planned registration count reaches break-even or better.")
```

4. Sample run guidance

Run the program as written to see a sample result. Replace the sample lists and constants with the values provided in your assignment. Keep the rule order and units consistent; for example, percentages should be entered as percentages and converted to fractions only in the calculation.